

Information to Industry regarding implementation of IS 1293

आईएस 1293 के अनुपालन के संबंध में उद्योग को जानकारी

IS 1293 : 2005 for Plugs and Socket-Outlets for Household and Similar Purposes of Rated Voltage up to and Including 250 Volt and Rated Current up to and Including 16 A is under BIS Certification.

250 वोल्ट तक के वोल्टेज और 16 एम्पियर तक के करंट वाले घरलू एवं इसी तरह के प्रयोजनों हतू प्लगों और सॉकेट-ऑऊटलेटों के लिए आई एस 1293 : 2005 बी आई एस प्रमाणन में हैं।

This Standard was revised in 2019 and published as IS 1293 : 2019. At present both 2005 and 2019 versions of IS 1293 are valid. They will remain valid till 23rd October 2023 after which 2005 version will be withdrawn and 2019 version shall remain in force. There are more than 390 BIS certification licences as per IS 1293.

यह मानक को 2019 में पुनरीक्षित किया गया था और आई एस 1293: 2019 के रूप में प्रकाशित किया गया था। वर्तमान में, आई एस 1293 के 2005 और 2019 दोनों संस्करण वैध हैं। ये दोनों दिनांक 23 अक्टूबर 2023 तक वैध रहेंगे, तत्पश्चात 2005 संस्करण वापिस लिया जाएगा और 2019 संस्करण प्रभावी रहेगा। आई एस 1293 के अंतर्गत 390 से अधिक बी आई एस प्रमाणन लाइसेंस हैं।

In June 2021, DPIIT notified Plugs and Socket-outlets as per IS 1293 for mandatory certification by BIS. This Quality Control Order is in effect from 01.06.2022.

जून 2021 में, डीपीआईआईटी ने बीआईएस द्वारा अनिवार्य प्रमाणन हेतु आई एस 1293 के अनुसार प्लगों एवं सॉकेट-आउटलेटों को अधिसूचित किया था। यह गुणता नियंत्रण आदेश दिनांक 01.06.2022 से प्रभावी है।

In view of the concerns raised by the Industry with respect to the coverage of products under the Standard and the QCO, a meeting was held earlier January 2021 with all stakeholders including Industry Associations. Based on the outcome of the meeting and feedback from the industry, this information was published earlier vide *Version II dated 25.02.2021* for clarity on coverage of products under IS 1293. The same is updated with current concurrent running date of IS 1293.

मानको के अंतर्गत उत्पादों की कवरेज तथा क्यूसीओ के संबंध में उद्योग द्वारा उठाए गए मामलों को देखते हुए, उद्योग संघों सहित सभी स्टॉकहोल्डर्स के साथ जनवरी 2021 में एक बैठक आयोजित की गई थी। इस बैठक के परिणामों तथा उद्योग से प्राप्त फीडबैक के आधार पर यह जानकारी आईएस 1293 के तहत उत्पादों की कवरेज पर स्पष्टता हेतु *Version II dated 25.02.2021* द्वारा पहले प्रकाशित की गई थी। इसे IS 1293 की वर्तमान समवर्ती चलने की तारीख के साथ अपडेट किया गया है।

The following Table 1 gives a list of Plugs and socket-outlets of different ratings and configurations that are covered in IS 1293: 2005 and IS 1293: 2019 versions and the varieties that are under discussion to be included in the Standard.

Table 1: Plugs and socket outlets covered in IS 1293

IS 1293: 2005 with 5 Amendments (valid upto 23.10.2021)	IS 1293: 2019	Amendment 1 to IS 1293: 2019 Implemented w.e.f 16.12.2020	Amendment 2 to IS 1293: 2019 – under discussion (varieties to be added)
2.5 A/2P (non-rewireable plugs only)	2.5 A/2P (non-rewireable plugs only)		
6 A/ 2P (Plugs only)	6 A/ 2P	6 A/ 2P (non-rewireable plugs only)	
6 A/2P+E	-	6A/2P+E	
10 A/2P+E	-	-	10 A/2P (non-rewireable plugs only) 10 A/2P+E
16A/2P+E	16A/2P+E	16 A/ 2P (non-rewireable plugs only)	
6/16 A or 10/16 A (Combined socket-outlets only)	6/16 A (Combined socket-outlets only)	6/16 A (Combined socket-outlets only) (explicitly added in Table 2)	6/16 A or 10/16 A (Combined socket-outlets only)

**all configurations are in round pins only*

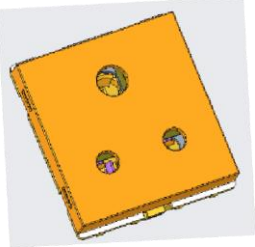
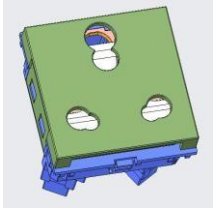
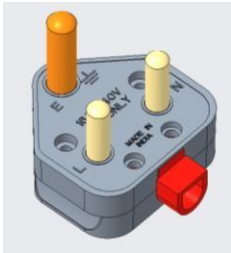
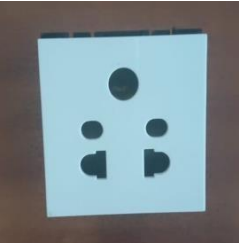
**2P (non-rewireable plugs) configuration for class II appliances only.*

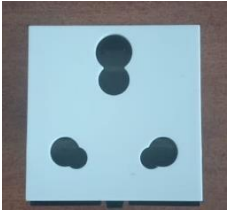
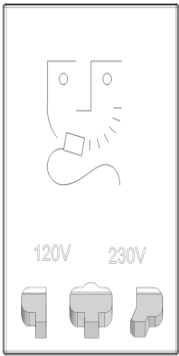
Any variety of Plug or Socket-outlet as per the ratings and configurations above are covered in the Standard and certification for the same shall be obtained from BIS. As on date manufacturers can opt for Licence as per either 2005 or 2019 version depending on the varieties manufactured by him.

At present, varieties like 2P, 16 A are not covered in either of the two versions of IS 1293. But it has been brought under 2019 version through Amendment 1. Amendment 1 has been recently notified, and now Licensees can get this included in the scope of their Licence.

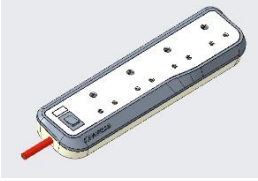
Table 2 below gives a list of different varieties with clarification on their coverage under the scope of IS 1293.

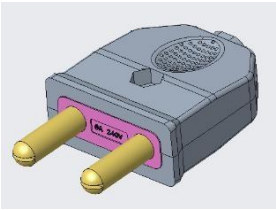
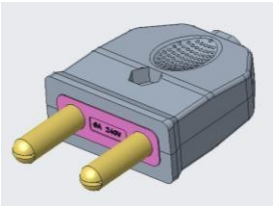
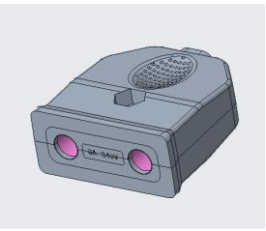
Table 2

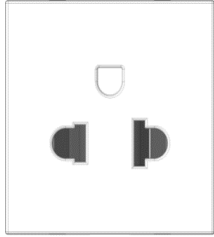
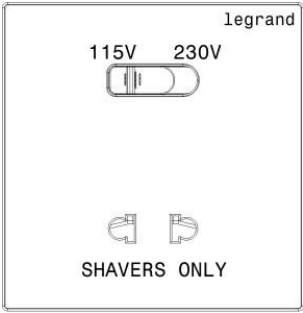
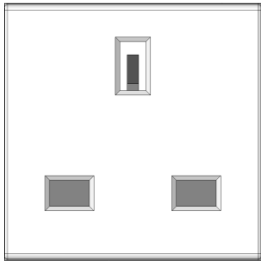
Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
1		6A, 3Pin Socket (2P+E)	Included in Scope	Included in Scope	
2		6/16A Socket (2P+E)	Included in Scope	Included in Scope	
3		6A & 16A Plugs (2P+E)	Included in Scope	Included in Scope	
4		Multistandard Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee is under process to explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
5		6A, 5pin Universal Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee is under process to explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

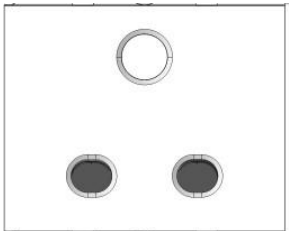
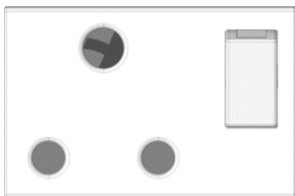
Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
6		5pin Round type Socket	Not Included in the scope	Not Included in the scope	Already deleted from the scope of IS 1293: 2019.
7		25A Socket, 10A/ 25A Socket	Not Included in the scope	Not Included in the scope	As per IS 1293: 2019, the requirements for 25 A rated accessories are under consideration.
8		25A Plug (2P+E)	Not Included in the scope	Not Included in the scope	As per IS 1293: 2019, the requirements for 25 A rated accessories are under consideration.
9		Shaver Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
10		Multiple Portable Socket Outlet (Cord Extension set with Multi Standard or Universal Sockets with & without USB)	Not Included in the scope	Not Included in the scope	IS 1293 applies to plugs and portable socket-outlets incorporated in cord extension sets. Particular requirements of the whole product is covered under separate standard which is under preparation.
11		10A, 16A plug for Class II Appliances	10A rating Included in the scope 16A rating is Included as 2P + ⊕	10 A rating not included	2P (non-rewireable plugs only)/ 3P 10 A- will be included through amendment no 2 which is under preparation and may take 8-10 months to get printed. 2P/16A (non-rewirable plugs only)- has already been included through amendment no 1 which has been notified. 2P+dummy earth pins for 6 & 16A, has already been discussed by the technical committee and it was decided not to include it in the standard.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
12	 	3-Pin Universal Socket (2P+E)	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee is under process to explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
13	 	Adaptors (Travel/Portable) 6A 2-PIN & 3PIN Adaptor 6/16A Multi socket Adaptor Multi standard 2pin/3pin Adaptors	Not Included in the scope	Not Included in the scope	The particular requirements for adaptors are covered under separate standard, which is under preparation
14		16A 2-pin Euro US socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
15		2-pin socket round	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
16		2.5 & 6A Plug portable (Rewirable)	6A plug is included	Not Included in the scope	2P/2.5 A- only non-rewireable plugs are permitted as per both versions. 2P/6 A - only non-rewirable plugs are allowed as per 2019 version + amendment 1.
17		2.5 & 6A Plug portable (Non Rewirable)	Included in the scope	Included in the scope	
18		2.5 & 6A Socket 2 Pin portable	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
19		PRCD (Protection from residual current device) Shock safe Plug	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
20		PRCD (Protection from residual current device) Shock safe Adaptor	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
21		16A 3 pin Euro US Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
22		Shaver Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
23		Shaver Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
24		13A 3 pin (2P +E) BS Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
25		6A, 2/3 pin socket (Round pin)	Not Included in the scope	Not Included in the scope	2P/6A sockets are not covered and will not be included in the standard. 16.5 mm pitch distance for 2.5 A plugs and socket-outlets has already been phased out from 31st December 2011. (refer amendment no. 5 to IS 1293: 2005). No subsequent changes in the pitch dimensions made in the revised IS 1293.
26		16A Switched Socket	Not Included in the scope	Included in the scope	

Information to Industry regarding implementation of IS 1293

IS 1293 के अनुपालन के संबंध में उद्योग को जानकारी

IS 1293: 2005 for Plugs and Socket-outlets for Household and Similar Purposes of Rated Voltage up to and Including 250 V and Rated Current up to and Including 16 A is under BIS Certification with more than 350 Licences.

250 वोल्ट तक के वोल्टेज और 16 एम्पीयर तक के करंट वाले घरेलू एवं इसी तरह के प्रयोजनों हेतु प्लगों और सॉकेट-आउटलेटों के लिए आईएस 1293:2005 है, इसके अंतर्गत 350 से अधिक बीआईएस प्रमाणन लाइसेंस हैं।

This Standard was revised in 2019 and published as IS 1293: 2019. At present both 2005 and 2019 versions of IS 1293 are valid. They will remain valid till 23rd October 2021 after which 2005 version will be withdrawn and 2019 version shall remain in force.

यह मानक 2019 में पुनरीक्षित किया गया था और आईएस 1293:2019 के रूप में प्रकाशित किया गया था। वर्तमान में, आईएस 1293 के 2005 और 2019 दोनों संस्करण वैध हैं। ये दोनों दिनांक 23 अक्टूबर 2021 तक वैध रहेंगे, तत्पश्चात् 2005 संस्करण वापिस लिया जाएगा और 2019 संस्करण प्रभावी रहेगा।

In December 2019, DPIIT notified Plugs and Socket-outlets as per IS 1293 for mandatory certification by BIS. This Quality Control Order is in effect from 01. 12. 2020.

दिसम्बर 2019 में, डीपीआईआईटी ने बीआईएस द्वारा अनिवार्य प्रमाणन हेतु आईएस 1293 के अनुसार प्लगों एवं सॉकेट-आउटलेटों को अधिसूचित किया था। यह गुणता नियंत्रण आदेश दिनांक 01.12.2020 से प्रभावी है।

In view of the concerns raised by the Industry with respect to the coverage of products under the Standard and the QCO, a meeting was held with all stakeholders including Industry Associations. Based on the outcome of the meeting and feedback from the industry, this information is being published for clarity on coverage of products under IS 1293.

मानकों के अंतर्गत उत्पादों की कवरेज तथा क्यूसीओ के संबंध में उद्योग द्वारा उठाए गए मामलों को देखते हुए उद्योग संघों सहित सभी स्टैकहोल्डर्स के साथ एक बैठक आयोजित की गई थी। इस बैठक के परिणामों तथा उद्योग से प्राप्त फीडबैक के आधार पर यह जानकारी आईएस 1293 के तहत उत्पादों की कवरेज पर स्पष्टता हेतु प्रकाशित की जा रही है।

The following Table 1 gives a list of Plugs and socket-outlets of different ratings and configurations that are covered in IS 1293: 2005 and IS 1293: 2019 versions and the varieties that are under discussion to be included in the Standard.

Table 1: Plugs and socket outlets covered in IS 1293

IS 1293: 2005 with 5 Amendments (valid upto 23.10.2021)	IS 1293: 2019	Amendment 1 to IS 1293: 2019 Implemented w.e.f 16.12.2020	Amendment 2 to IS 1293: 2019 – under discussion (varieties to be added)
2.5 A/2P (non-rewireable plugs only)	2.5 A/2P (non-rewireable plugs only)		
6 A/ 2P (Plugs only)	6 A/ 2P	6 A/ 2P (non-rewireable plugs only)	
6 A/2P+E	-	6A/2P+E	
10 A/2P+E	-	-	10 A/2P (non-rewireable plugs only) 10 A/2P+E
16A/2P+E	16A/2P+E	16 A/ 2P (non-rewireable plugs only)	
6/16 A or 10/16 A (Combined socket-outlets only)	6/16 A (Combined socket-outlets only)	6/16 A (Combined socket-outlets only) (explicitly added in Table 2)	6/16 A or 10/16 A (Combined socket-outlets only)

**all configurations are in round pins only*

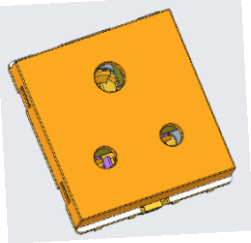
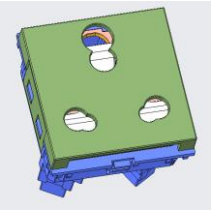
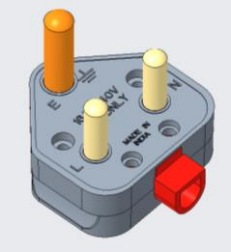
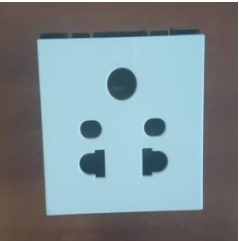
**2P (non-rewireable plugs) configuration for class II appliances only.*

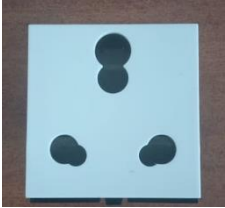
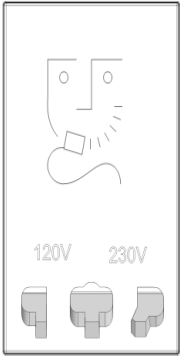
Any variety of Plug or Socket-outlet as per the ratings and configurations above are covered in the Standard and certification for the same shall be obtained from BIS. As on date manufacturers can opt for Licence as per either 2005 or 2019 version depending on the varieties manufactured by him.

At present, varieties like 2P, 16 A are not covered in either of the two versions of IS 1293. But it has been brought under 2019 version through Amendment 1. Amendment 1 has been recently notified, and now Licensees can get this included in the scope of their Licence.

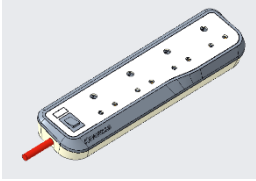
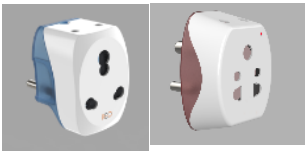
Table 2 below gives a list of different varieties with clarification on their coverage under the scope of IS 1293.

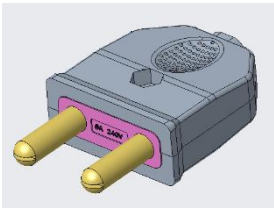
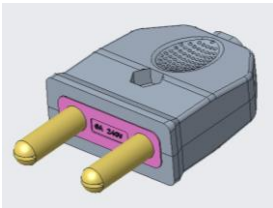
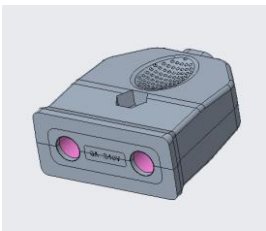
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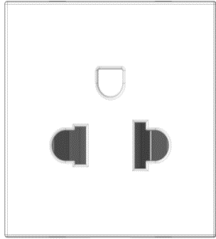
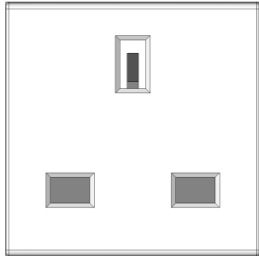
Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
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2		6/16A Socket (2P+E)	Included in Scope	Included in Scope	
3		6A & 16A Plugs (2P+E)	Included in Scope	Included in Scope	
4		Multistandard Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee is under process to explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
5		6A, 5pin Universal Socket	Not Included in the scope	Not Included in the scope	Already deleted from the scope of IS 1293: 2019.

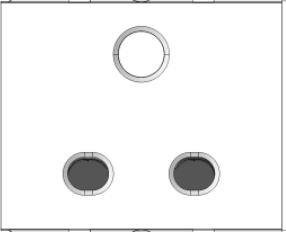
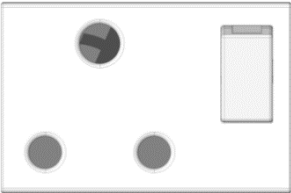
Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
6		25A Socket, 10A/ 25A Socket	Not Included in the scope	Not Included in the scope	As per IS 1293: 2019, the requirements for 25 A rated accessories are under consideration.
7		25A Plug (2P+E)	Not Included in the scope	Not Included in the scope	As per IS 1293: 2019, the requirements for 25 A rated accessories are under consideration.
8		Shaver Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
9	  	Multiple Portable Socket Outlet (Cord Extension set with Multi Standard or Universal Sockets with & without USB)	Not Included in the scope	Not Included in the scope	IS 1293 applies to plugs and portable socket-outlets incorporated in cord extension sets. Particular requirements of the whole product is covered under separate standard which is under preparation.
10		10A, 16A plug for Class II Appliances	10A rating Included in the scope 16A rating is Included as 2P + 	10 A rating not included	2P (non-rewireable plugs only)/ 3P 10 A- will be included through amendment no 2 which is under preparation and may take 8-10 months to get printed. 2P/16A (non-rewirable plugs only)- has already been included through amendment no 1 which has been notified. 2P+dummy earth pins for 6 & 16A, has already been discussed by the technical committee and it was decided not to include it in the standard.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
11	 	3-Pin Universal Socket (2P+E)	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee is under process to explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
12	 	Adaptors (Travel/Portable) 6A 2-PIN & 3PIN Adaptor 6/16A Multi socket Adaptor Multi standard 2pin/3pin Adaptors	Not Included in the scope	Not Included in the scope	The particular requirements for adaptors are covered under separate standard, which is under preparation
13		16A 2-pin Euro US socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
14		2-pin socket round	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
15		2.5 & 6A Plug portable (Rewirable)	6A plug is included	Not Included in the scope	2P/2.5 A- only non-rewireable plugs are permitted as per both versions. 2P/6 A - only non-rewirable plugs are allowed as per 2019 version + amendment 1.
16		2.5 & 6A Plug portable (Non Rewirable)	Included in the scope	Included in the scope	
17		2.5 & 6A Socket 2 Pin portable	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
18		PRCD (Protection from residual current device) Shock safe Plug	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
19		PRCD (Protection from residual current device) Shock safe Adaptor	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
20		16A 3 pin Euro US Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
21		Shaver Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
22		Shaver Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.
23		13A 3 pin (2P +E) BS Socket	Not Included in the scope	Not Included in the scope	Not covered in the standard. The technical committee will explore the possibility whether technical specification/ Indian standard can be formulated for the same or not.

Sr. No.	Ref. image	Product Description & Rating	IS 1293 :2005	IS 1293 :2019	Remarks
24		6A, 2/3 pin socket (Round pin)	Not Included in the scope	Not Included in the scope	2P/6A sockets are not covered and will not be included in the standard. 16.5 mm pitch distance for 2.5 A plugs and socket-outlets has already been phased out from 31st December 2011. (refer amendment no. 5 to IS 1293: 2005). No subsequent changes in the pitch dimensions made in the revised IS 1293.
25		16A Switched Socket	Not Included in the scope	Included in the scope	

PRODUCT MANUAL FOR POLYVINYL CHLORIDE INSULATED UNSHEATHED AND SHEATHED CABLES/CORDS WITH RIGID AND FLEXIBLE CONDUCTOR FOR RATED VOLTAGES UP TO AND INCLUDING 1100 V ACCORDING TO IS 694: 2010

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

1.	Product	:	IS 694: 2010	
	Title	:	Polyvinyl Chloride Insulated Unsheathed And Sheathed Cables/Cords With Rigid And Flexible Conductor For Rated Voltages Up To And Including 1100 V	
	No. of Amendments	:	4	
2.	Sampling Guidelines:			
a)	Raw material	:	Annealed, bare or tinned high conductivity copper wires Aluminium wires	IS 8130
b)	Grouping guidelines	:	Please refer ANNEX – A	
c)	Sample Size	:	(i) 5 meters Copper/Aluminium wire (before stranding) (ii) Cu wire – 100 g (for Cu purity test) (ii) 25 meters PVC Insulated Cable/ Cord	
3.	List of Test Equipment	:	Please refer ANNEX – B .	
4.	Scheme of Inspection and Testing	:	Please refer ANNEX – C .	
5.	Possible tests in a day	:	Please refer ANNEX - D	
6.	Scope of the Licence : PVC Insulated, Unsheathed/Sheathed, Circular/Flat, Cables/Cords with Rigid/Flexible Copper (Bare/Tinned)/Aluminum Conductor, Class 1/2/5, Insulation Type A/C/D, Sheath Type ST1/ST2/ST3, Category 01/02/FR/FR-LSH, for working Voltages upto and including 1100 V, for the following varieties: a) Single Core for Sizes upto and including b) Multi-core upto and including Cores, Sizes upto and including			

ANNEX A**Grouping Guidelines**

I. 1(a). Samples of each variety of Cable shall be tested considering the following:

- i. Material for Conductor – Aluminium, Copper
- ii. Type of Conductor – Rigid (Class 1, 2) , Flexible (Class 5)
- iii. No. of Cores (Single Core, Multi-core)
- iv. Type of Insulation (A, C, D)
- v. Type of Sheath (ST1, ST2, ST3)
- vi. Category of Cable (01, 02, FR ,FR-LSH)

1(b). The following relaxation may be given when a variety is tested for all the requirements:

Variety Tested	Additional Variety that may be covered
Tinned Copper	Bare Copper
Stranded (Class 2)	Solid (Class 1)
Multi-core	Single Core
Type A Insulation	Type D Insulation
Sheathed	Unsheathed
Type ST1 Sheath	Type ST3 Sheath
(a) FR-LSH	(a) FR and 01
(b) FR	(b) 01
(c) 02	(c) 01
Flat Cable	Circular Cable

2. The Firm shall declare the Varieties and Sizes of various Cables they intend to cover in the Licence. Cable with any Size (Nominal Cross Sectional Area of Conductor) and no. of Cores, preferably the largest, intended to be covered in the Licence may be drawn for Testing. The Scope of Licence may be restricted based on the Manufacturing and Testing capabilities of the Manufacturer.
 3. During the operation of the Licence, BO shall ensure that all the Varieties covered in the Licence are tested in rotation to the extent possible.
- II. A typical example for drawal of samples to cover the entire Varieties under the Scope of the Licence is given below for the purpose of general guidance:
- (a) Sample 1: Three Core Cable, Rigid Aluminium Stranded Conductor (Class 2), Insulation Type A, Sheath Type ST1, Category 02.
- and*
- (b) Sample 2: Two Core Flat Cable, Flexible tinned Copper Conductor (Class 5), Insulation Type C, Sheath Type ST2, Category FR-LSH.

ANNEX B**List Of Test Equipment***Major test equipment required to test as per the Indian Standard*

Sl. No.	Test Equipment	Tests used in with Clause Reference	
		Cl. Ref.	Tests
1	Vernier Callipers	4, 17.1.1, 5.3, 8.3, 9	Measurement of thickness & Outer Dia.
2	Steel Scale	10.4, 4.1, 5.4	Measurement of Flammability, Annealing and Elongation
3	Micrometer	4, 17.1.1	Measurement of Wire Dia.
4	Measuring Microscope	4, 17.1.1, 10	Measurement of thickness (insulation & sheath) and hot deformation test
5	Graduated Magnifying Glass	4, 17.1.1, 10	Measurement of thickness (insulation & sheath) and hot deformation test
6	Tensile Testing Machine	10	Tensile Strength, Elongation Test & Annealing Test
7	Dumb-Bell Cutting Machine with Die	10	Tensile Strength & Elongation Test
8	Hot Air oven with thermostatic Temperature controller	10	Heat Shock, Hot Deformation & Shrinkage Test
9	Thermal Ageing Oven with Thermostatic Temperature controller, Air flow Meter & Hour Meter	10	Tensile Strength & Elongation After ageing, Loss Of Mass Test
10	Water Bath with thermostatic temperature controller, Stirrer & Hour Meter	10.1, 10.2	Water immersion A.C. & D.C. Test, Insulation Resistance Constant/ Volume Resistivity Test
11	Megohmmeter	10.1	Insulation Resistance Constant/ Volume Resistivity Test
12	Heat Shock Mandrels	10	Heat Shock Test

13	Smooth metal Mandrels	10	Cold Bend Test
14	Low temperature Impact Apparatus	10	Cold Impact Test
15	Hot Pressure Test Apparatus with Weights	10	Hot Deformation Test
16	Physical Weighing Balance with Weights	4, 17.1.1, 10	Loss of Mass Test & Thermal Stability Test, Persulphate Test, Flammability Test
17	Kelvin Double Bridge with galvanometer, D.C. Source	10	Conductor Resistance Test
18	Flammability Test apparatus with burner, Gas cylinder, Scale, Stopwatch & 0.71 mm copper wire	10.4	Flammability Test
19	Conditioning chamber & Refrigerator with Humidity Indicator	10	Tensile Strength, Elongation Test, Cold Bend, Cold Impact & Thermal stability Test
20	AC Spark Tester	10.3	Spark Testing
21	D.C. High Voltage Test set	10.1	D.C. High Voltage Test
22	A C. High Voltage Test set	10.1, 10.2	A.C. High Voltage Test
23	Megohm Box	10.1	Insulation Resistance Test
24	Glass Thermometer	10	Conductor Resistance & Insulation Resistance Test
25	Balance Digital	10.4	Flammability Test
26	Stop Watch	10.4	Flammability Test
27	Thermal stability test apparatus with pH paper & tubes	10	Thermal Stability Test
Additional Testing Equipment Required for Testing of Category (FR), (FR-LSH)			
1	Oxygen Index test apparatus, Oxygen & Nitrogen gas	10	Oxygen Index and Temperature Index Test

2	Tube Furnace, Quartz Combustion Tube, Porcelain combustion boat, Wash Bottles, Glass Tubing and Silicon Rubber Stoppers, Air Flow Meter, Reagents	10	Halogen Acid Gas Evolution Testing
3	Smoke density test apparatus	10	Smoke density rating test

The above lists are indicative only and may not be treated as exhaustive.

ANNEX C

Scheme Of Inspection And Testing

1. LABORATORY - A laboratory shall be maintained which shall be suitably equipped (as per the requirement given in column 2 of Table 1) and staffed, where different tests given in the specification shall be carried out in accordance with the methods given in the specification.

1.1 The manufacturer shall prepare a calibration plan for the test equipment.

2. TEST RECORDS – The manufacturer shall maintain test records for the tests carried out to establish conformity.

3. LABELLING AND MARKING – In addition to the requirements of IS 694: 2010, Identification in code or otherwise shall be either stencilled on reel/drum, packages of individual or contained in the label attached to the coil, in order that the date of manufacture and control unit can be traced back to factory records.

4. CONTROL UNIT – A cable/ cord of continuous length manufactured/ extruded under similar conditions of production for one nominal cross-sectional area and class of conductor shall constitute a control unit.

5. LEVELS OF CONTROL - The tests as indicated in column 1 of [Table 1](#) and the levels of control in column 3 of [Table 1](#), shall be carried out on the whole production of the factory which is covered by this plan and appropriate records maintained in accordance with paragraph 2 above.

6. REJECTIONS – Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1

(1)						(2)	(3)		
Test Details						Test equipment requirement R: required (or) S: Sub-contracting permitted	Levels of Control		
Cl.	Requirement	Test		Test Methods			No. of Sample	Frequency	Remarks
		Clause	Ref	Part	Ref				
4.1 & 4.2	Conductor Material					R	One	Each Coil/bobbin of finished wire drawn or received	No further testing is required if accompanied with Test Certificate or ISI marked.
	a) Resistance	7.3	IS 8130	5	IS 10810				
	b) Tensile strength (for Al, before stranding)	7.2.1		2					
	c) Annealing test (before stranding)	7.1.2.1, 7.2.3		1					
	d) Wrapping (for Al)	7.2.2		3					
	e) Purity test (for Cu)	Table 4	IS 191	-	-	S		For each consignment	
10	a) Conductor Resistance	7.3	IS 8130	5	IS 10810	R	Each Coil	Entire length of core after extrusion of insulation	If so desired these routine tests may be conducted on the entire length of the control unit before cutting into individual coils.
	a) High Voltage at room temperature <i>or</i>	10.2	IS 694	45					
	b) Spark test (as applicable)	10.3		44					
5.1, 6, 7, 8.2 11 18.1.3, 19.1.4, 20.1.3, 21.1.3, 22.1.3	a) Material b) Identification c) Assembly of cores	Section 2 & 3	IS 694			R	Each Control Unit	Visual examination	

5.3	a) Thickness of Insulation	Section 2 & 3						Each Control Unit of 10,000 m or less	
8.3	b) Thickness of Sheath	Sec 3	IS 694	6	IS 10810	R	Two		
9	c) Overall dimension	Section 2 & 3		-	-				
9.1	Ovality	Sec 3	IS 694			R	Two		
4.1	a) Tensile strength (for Al, after stranding)	7.2.1	IS 8130	2	IS 10810	R	Two		
	a) Wrapping test (for Al)	7.2.2		3					
4.2	Persulphate test	7.1.1	IS 8130	4	IS 10810	S	One	Once in a month	For tinned Copper Only
10	Insulation	Table 1	IS 5831	7	IS 10810	R	One	Cable/Cord of each size and type manufactured in a month from one consignment of PVC compound	#
	a) Tensile strength b) Elongation at break Sheath a) Tensile strength b) Elongation at break	Table 2							#
10	Insulation Resistance/ Volume Resistivity	Table 1	IS 5831	43	IS 10810	R	One		Either IR Constant or Volume Resistivity to be recorded
10	Insulation & Sheath a) Heat Shock b) Hot deformation c) Loss of Mass d) Tests after Ageing e) Thermal Stability f) Shrinkage	Table 1 & 2	IS 5831	14 15 10 11 60 12	IS 10810	R	One		#

10	High Voltage (Water Immersion) Test	10.1	IS 694	45	IS 10810	R	One	Cable/Cord of each size and type manufactured in a month from one consignment of PVC compound	
10	Flammability test	10.4	IS 694	53	IS 10810	R	One	Cables/ Cord manufactured from each consignment of PVC compound	
10	Cold Bend test Cold Impact test	Table 1 & 2	IS 5831	20 21	IS 10810	R	One	Every month for each size and type from one consignment of PVC compound	
10	Additional Ageing test	10.9	IS 694	-	IS 10810	R	One		For Category 02 cable only
	Oxygen Index test	10.5		58					For FR & FR-LSH
	Temperature Index test	10.7		64					For FR & FR-LSH
	Halogen Acid Gas evaluation	10.6		59					For FR-LSH
	Smoke density rating	10.8		IS 13360 (Part 6/Sec 9)				For FR-LSH	

In case of failure, marking of that type and size shall be stopped and reasons for failure shall be investigated. Marking shall be done only if the sample from the control unit of improved quality meets the specified requirements. The original frequency shall be restored only after five consecutive samples have passed the requirements.

Note-1: Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empanelled by the Bureau.

Note-2: Levels of control given in column 3 are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control in column 3 with proper justification for approval by BO Head.

ANNEX D

Possible Tests in a day

Dimensions

- Maximum Conductor dia.
- Overall dimensions
- Ovality
- Thickness of sheath and insulation

Annealing Test (for Copper wires)

Tensile Test (for Al wires)

Wrapping test (for Al wires)

Conductor Resistance

Tensile Test (before ageing) on Insulation and Sheath

Shrinkage Test

Hot Deformation

Heat Shock Test

Insulation Resistance Test

HV Test (at Room Temperature)

Flammability Test



उत्पाद मैनुअल
आई एस 374 : 2019 के अनुसार
विद्युत् छत के प्रकार के पंखों - विशिष्टि (चौथा पुनरीक्षण) के लिए
दस्तावेज संख्या - पीएम/आईएस 374/2/ जनवरी 2023

भारतीय मानक ब्यूरो की स्कीम-I (अनुरूपता मूल्यांकन) विनियम, 2018 के तहत यह उत्पाद मैनुअल प्रमाणीकरण के प्रचालन में रीति और पारिश्रिता की सुसंगतता सुनिश्चित करने के लिए सभी क्षेत्रीय/शाखा कार्यालयों और लाइसेंसी द्वारा संदर्भ सामग्री के रूप में उपयोग किया जाएगा। बीआईएस प्रमाणीकरण लाइसेंस/प्रमाणपत्र प्राप्त करने के इच्छुक भावी आवेदकों द्वारा भी इस दस्तावेज का उपयोग किया जा सकता है।

**PRODUCT MANUAL FOR
ELECTRIC CEILING TYPE FANS —
SPECIFICATION (FOURTH REVISION)
ACCORDING TO IS 374 : 2019**

Document No - PM/IS 374/2/January 2023

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, ९, बहादुर शाह ज़फ़र मार्ग
Manak Bhawan, 9, Bahadur Shah Zafar Marg
नई दिल्ली- ११०००२
New Delhi – 110002



PRODUCT MANUAL FOR

ELECTRIC CEILING TYPE FANS ACCORDING TO IS 374:2019

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

1.	Product	:	IS 374:2019
	Title	:	Electric Ceiling Type Fans
	No. of Amendments	:	1
2.	Sampling Guidelines:		
a)	Raw material	:	- Stampings as per IS 648 - Electronic Type Fan Regulators (if provided) as per IS 11037 - Remote Control Device (if provided) as per IS 14700 (Part 3/Sec 2)
b)	Grouping guidelines	:	Please refer ANNEX – A
c)	Sample Size	:	3 nos.
3.	List of Test Equipment	:	Please refer ANNEX – B
4.	Scheme of Inspection and Testing	:	Please refer ANNEX – C
5.	Possible tests in a day	:	Please refer ANNEX – D
6.	Scope of the Licence :		
	Licence is granted to use Standard Mark as per IS 374:2019 with the following scope:		
	Electric ceiling type fans, driven by squirrel cage induction motor (capacitor type) / Brushless dc motor fans, with or without Speed Regulators / Remote Control Device, for fan size(s)mm,V, 50 Hz, Insulation Class ... Rating Input Wattage ... Model No....		

BUREAU OF INDIAN STANDARDS
Manak Bhawan, 9, Bahadur Shah Zafar Marg,
New Delhi – 110002

ANNEX A

1. IS 374:2019 covers Electric ceiling fans driven by squirrel cage induction motor (capacitor type) as well as brushless dc motor fans with or without speed regulators/remote control device.
2. Manufacturer shall declare the type, fan size, input voltage, insulation class, and rated input wattage rating produced by them with a model number.
3. For grant of licence/ change in scope of licence, each model of fan to be tested. Change in aesthetics, i.e colour, visual design etc may not be treated as different model.
4. However, following relaxation in testing is given
 - i. If fans with speed regulator / remote control device is tested for one model, fans of the other model without speedregulator/remote control device may also be covered subject to the condition that fan size, input voltage, insulation class, and rated input wattage of both models are same.
 - ii. If the model already covered in the License is without speed regulator/ remote control device, fans of the other model with speed regulator/ remote control device may be covered subject to conformance to Cl. 10.1 to Cl. 10.8 of IS 374:2019 subject to the condition that fan size, input voltage, insulation class, and rated input wattage of both models are same.
5. The Firm shall declare the varieties of various electric ceiling fans they intend to cover in the Licence. The Scope of Licence may be restricted based on the Manufacturing capability and Testing facilities of the Manufacturer.
6. During the operation of the Licence, BO shall ensure that all the Varieties covered in the Licence are tested in rotation to the extent possible.
7. Manufacturer shall declare the type/ designation etc of accessories (regulator, remote control) compatible with the fan.

ANNEX-B**List Of Test Equipment***Major test equipment required to test as per the Indian Standard*

Sl. No	Test Equipment	Tests Used in with Clause reference
1	Voltmeter, Finger Test Apparatus	Cl. 8 of IS 302-2-80
2	Voltmeter, Ammeter, Wattmeter	Cl. 10 of IS 302-2-80
3	Test Corner consisting of dull-black painted plywood, Temperature Controller	Cl. 11 of IS 302-2-80
4	IR Meter, Micro ammeter, HV Tester, Stopwatch	Cl. 13 of IS 302-2-80
5	Humidity Test Chamber	Cl. 15 of IS 302-2-80
6	Metal foil, HV Test Supply	Cl. 16 of IS 302-2-80
7	Test Corner, Felt Strips, Copper/brass disks	Cl. 19 of IS 302-2-80
8	Inclined Test Apparatus	Cl. 20 of IS 302-2-80
9	Spring Hammer, Finger Test Apparatus	Cl. 21 of IS 302-2-80
10	Micrometer, VAW meter, Pressure Test Apparatus	Cl. 22 of IS 302-2-80
11	Flexing Test Apparatus	Cl. 23 of IS 302-2-80
12	Cord Grip Test Apparatus, Measuring Tape, Micro ohmmeter	Cl. 25 of IS 302-2-80
13	Torque Screwdriver	Cl. 26 of IS 302-2-80
14	Earth Contact Resistance Testing Panel (ECR Measurement), Voltmeter, Ammeter	Cl. 27 of IS 302-2-80
15	Torque Wrench	Cl. 28 of IS 302-2-80
16	Filler Gauge, Vernier Caliper, Micrometer	Cl. 29 of IS 302-2-80
17	Glow Wire Test Apparatus, Oven, Ball Pressure Apparatus, Heating Oven with temperature controller	Cl. 30 of IS 302-2-80
18	Humidity Chamber with temperature and humidity controller, measuring cylinder, Ammonium chloride, Carbon	Cl. 31 of IS 302-2-80
19	Tachometer, Power Factor Meter	Cl. 14.4, Cl. 14.5 of IS 374
20	Analogue Type Vane Anemometer/Digital Type Vane Anemometer	Cl. 14.3 of IS 374

The above list is indicative only and may not be treated as exhaustive.

ANNEX C

Scheme Of Inspection And Testing

1. LABORATORY- A laboratory shall be maintained which shall be suitably equipped (as per the requirement given in column 2 of Table 1) and staffed, where different tests given in the specification shall be carried out in accordance with the methods given in the specification.

1.1 The manufacturer shall prepare a calibration plan for the test equipment.

2. TEST RECORDS - The manufacturer shall maintain test records for the tests carried out to establish conformity.

3. LABELLING AND MARKING - As per the requirements of IS 374 : 2019.

4. LEVELS OF CONTROL - The tests as indicated in column 1 of Table 1 and the levels of control in column 3 of Table 1, shall be carried out on the whole production of the factory which is covered by this plan and appropriate records maintained in accordance with paragraph 2 above.

5. CONTROL UNIT – All ceiling fans of one model no. manufactured in a day shall constitute a control unit.

6. REJECTIONS - Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1

(1)				(2)	(3)		
Test Details				Test equipment requirement R: required (or) S: Sub-contracting permitted	Levels of Control		
Cl.	Requirement	Test Methods			No. of Sample	Frequency	Remarks
		Clause	Reference				
18.4	Earth Continuity Test	A-1	IS 302-1	R	Every Fan	—	
	Electric Strength Test	A-2	IS 302-1	R			
	Functional test	A-3	IS 302-1	R			
	Simple Running Test	18.4(d)	IS 374	R			
4	General Requirements						
4.2	Stampings	4.2	IS 374	S	Each consignment	*	
			IS 648				
4.3	Blades and Motor	4.3	IS 374	R	Each Fan		
4.6	Enclosure	4.6	IS 374	R	Each Motor		
9	Safety Requirements						
	Protection against access to live parts	8	IS 302-2-80	R	Three	Every control Unit	
	Power Input and Current	10	IS 302-2-80	R			
	Heating	11	IS 302-2-80	R			
	Leakage Current & Electric Strength at Operating Temperature	13	IS 302-2-80	R			
	Transient Overvoltage	14	IS 302-2-80	S	Three	Every year for each model no.	
	Moisture Resistance	15	IS 302-2-80	R	Three	Every week for each model no.	
	Leakage Current & Electric Strength	16	IS 302-2-80	R			
	Overload Protection of Transformers and Associated Circuits	17	IS 302-2-80	S	Three	Once a month for each model no.	
	Abnormal Operation	19	IS 302-2-80	S	Three	Every six months for each model no.	
	Stability & Mechanical Hazards	20	IS 302-2-80	S			

9	Mechanical Strength	21	IS 302-2-80	S	Three	Once a month for each model no.	
	Construction	22	IS 302-2-80	S		Every six months for each model no.	
	Internal Wiring	23	IS 302-2-80	S			
	Components	24	IS 302-2-80	S			
	Supply Connection & External Flexible Cords	25	IS 302-2-80	S			
	Terminals for External Conductors	26	IS 302-2-80	S			
	Provision for Earthing	27	IS 302-2-80	R	Every fan		
	Screws and Connections	28	IS 302-2-80	R	Three	Every six months for each model no.	
	Clearances, Creepage Distances & Solid Insulation	29	IS 302-2-80	R			
	Resistance to Heat and Fire	30	IS 302-2-80	S			
	Resistance to Rusting	31	IS 302-2-80	S			
	Radiation, Toxicity & Similar Hazards	32	IS 302-2-80	S			
15	Performance Requirements	14.3, 15	IS 374	R	Three	Each control unit	
14.4& 14.5	Speed and Power Factor	14.4& 14.5	IS 374	R			
10.1 to 10.6,10.8	Speed Regulator	10.1 to 10.6,10.8	IS 374	R	Once a week for each model no.		
10.7	Speed Regulator	10.7	IS 374	S	Each consignment		*
			IS 11037, IS 14700 (Part 3/Sec 2)				
11	Starting	11	IS 374	R	Three	Each control unit	
12	Interchangeability	12	IS 374	R			
13	Silent Operation	13	IS 374	R			

17	Test for Harmonic Distortion	17	IS 374	S	Three	Every six months for each model no.	
16	Endurance Test	16	IS 374	S	Three	Once a month for each model no.	

* No further testing is required if accompanied with Test certificate or ISI Marked.

Note -1: Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empaneled by the Bureau

Note- 2: Levels of control given in column 3 are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control in column 3 with proper justification for approval by BO Head.

ANNEX D
Possible Tests in a day

1. Earth Continuity Test (A-1)
2. Electric Strength Test (A-2)
3. Functional test (A-3)
4. Simple Running Test (Cl. 18.4 d)
5. Protection against access to live parts (Cl. 8)
6. Power Input and Current (Cl. 10)
7. Leakage Current and Electric Strength at operating temperature (Cl. 13 of IS 302-2-80)
8. Provision for Earthing (Cl. 27)
9. Starting (Cl. 11)
10. Interchangeability (Cl. 12)
11. Silent Operation (Cl. 13)
12. Performance Requirements (Cl. 14.3, 15)
13. Speed and Power Factor (14.4 and 14.5)



उत्पाद मैनुअल

आई एस 1293 : 2019 के अनुसार

घरेलु और ऐसे ही सामान्य प्रयोजनों के लिए 250 वोल्ट तक की रेटित वोल्टता वाले और 16 एम्पीयर तक की रेटित करंट वाले प्लग और सॉकेट- निकास (चौथा पुनरीक्षण) के लिए

दस्तावेज संख्या – पीएम/आईएस 1293/3/ मई 2024

विभिन्न उत्पादों के लिए भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम, 2018 की योजना- I के तहत प्रमाणन के संचालन में एकरूपता और पारदर्शिता के लिए इस उत्पाद मैनुअल का उपयोग सभी क्षेत्रीय / शाखा कार्यालयों और लाइसेंसधारियों द्वारा संदर्भ सामग्री के रूप में किया जाएगा। दस्तावेज का उपयोग बीआईएस प्रमाणन प्राप्त करने के इच्छुक संभावित आवेदकों द्वारा भी किया जा सकता है।

PRODUCT MANUAL FOR

Plugs and Socket-Outlets for Household and Similar Purposes of Rated Voltage up to and Including 250 V and Rated Current up to and Including 16 A — Specification (Fourth Revision)

ACCORDING TO IS 1293 : 2019

Document No - PM/IS 1293/3/May 2024

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure uniformity of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification.

भारतीय मानक ब्यूरो

BUREAU OF INDIAN STANDARDS

मानक भवन, ९, बहादुर शाह ज़फ़र मार्ग

Manak Bhawan, 9, Bahadur Shah Zafar Marg

नई दिल्ली- ११०००२

New Delhi – 110002



उत्पाद मैनुअल

घरेलु और ऐसे ही सामान्य प्रयोजनों के लिए 250 वोल्ट तक की रेटित वोल्टता वाले और 16 एम्पीयर तक की रेटित करंट वाले प्लग और सॉकेट- निकास (चौथा पुनरीक्षण)
आई एस 1293 : 2019 के अनुसार

PRODUCT MANUAL

Plugs and Socket-Outlets for Household and Similar Purposes of Rated Voltage up to and Including 250 V and Rated Current up to and Including 16 A — Specification (Fourth Revision)
ACCORDING TO IS 1293 : 2019

विभिन्न उत्पादों के लिए भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम, 2018 की योजना- I के तहत प्रमाणन के संचालन में एकरूपता और पारदर्शिता के लिए इस उत्पाद मैनुअल का उपयोग सभी क्षेत्रीय / शाखा कार्यालयों और लाइसेंसधारियों द्वारा संदर्भ सामग्री के रूप में किया जाएगा। दस्तावेज़ का उपयोग बीआईएस प्रमाणन प्राप्त करने के इच्छुक संभावित आवेदकों द्वारा भी किया जा सकता है।

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure uniformity of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification.

1.	मानक संख्या IS No.	:	IS 1293 : 2019
	शीर्षक Title	:	Plugs and Socket-Outlets for Household and Similar Purposes of Rated Voltage up to and Including 250 V and Rated Current up to and Including 16 A — Specification (Fourth Revision)
	संशोधनों की संख्या No. of amendments	:	2
2.	नमूना दिशानिर्देश Sampling Guidelines:		
a)	कच्चा माल Raw material	:	As per the requirements of relevant clauses, as applicable, in case the components are under mandatory certification, they shall bear

			BIS Standard Mark
b)	समूहीकरण दिशानिर्देश Grouping Guidelines	:	Please refer ANNEX – A
c)	नमूने का परिमाण Sample Quantity	:	10 numbers for Socket Outlets and 15 numbers for Plugs
3.	परीक्षण उपकरणों की सूची List of Test Equipment	:	Please refer ANNEX – B
4.	निरीक्षण और परीक्षण की स्कीम Scheme of Inspection and Testing	:	Please refer ANNEX – C
5.	एक दिन में संभावित परीक्षण Possible tests in a day:	:	Please refer ANNEX-D
6.	लाइसेंस का दायरा /Scope of the Licence	:	Please refer ANNEX-E
7.	अन्य दिशानिर्देश/ Any other guidelines	:	Please refer ANNEX-F for guidelines for outsourcing of moulded bodies

ANNEX – A

समूहीकरण दिशानिर्देश **Grouping Guidelines**

1. The following aspects of a Plug or Socket-Outlet shall be taken into consideration for grouping of different varieties for the purpose of GoL/ CSoL:
 - a. Current rating (as per CI 6 of IS 1293);
 - b. Voltage rating (as per CI 6 of IS 1293);
 - c. Degree of protection against access to hazardous parts and against harmful ingress of solid foreign objects (as per CI 7.1.1 of IS 1293);
 - d. Degree of protection against harmful ingress of water (as per CI 7.1.2 of IS 1293);
 - e. Provision for earthing (as per CI 7.1.3 of IS 1293);
 - f. Method of connecting the cable (as per CI 7.1.4 of IS 1293);
 - g. Type of terminals (as per CI 7.1.5 of IS 1293);
 - h. Method of application/ mounting of the socket-outlet (as per CI 7.2.2 of IS 1293);
 - i. Intended use, if any (as per CI 7.2.4 of IS 1293);
 - j. Intended Class of appliance (as per CI 7.3 of IS 1293);
 - k. Type of Socket-Outlet (Fixed/ Portable/Combined/ multiple portable socket-outlets for Appliances);
 - l. Degree of protection against electric shock (as per CI 7.2.1 of IS 1293);
 - m. Existence of shutters (as per CI 7.2.3 of IS 1293);
 - n. Plug pins with insulating sleeves
2. At least one plug and socket outlet of each current rating and combination thereof shall be tested to cover the entire range.
3. If a particular voltage rating is tested, lower ratings may be covered.
4. Additionally, the following relaxations may be permitted when a variety is tested for all the requirements:
 - a. If accessories with a particular IP is tested, lower IPs may also be covered.
 - b. For the parameters mentioned at 1.1(e) to 1.1(k) above, at least one plug or socket-outlet (as applicable) of each aspect shall be tested to cover the entire range. However, if accessories with screwless terminals for rigid and flexible conductors are tested, accessories with screwless terminals for rigid conductors shall also be covered.
 - c. If socket-outlets with increased protection are tested, socket-outlets with normal

protection are also covered.

- d. If socket-outlets with shutters are tested, socket-outlets without shutters are also covered.
 - e. If plugs having pins with insulating sleeves are tested, plugs having pins without insulating sleeves are also covered.
5. The Firm shall declare the varieties of Plugs or Socket-outlets intended to be covered in the Licence. The Scope of Licence may be restricted based on the manufacturing and testing capabilities of the manufacturer.
 6. During the operation of the Licence, BO shall ensure that all the varieties covered in the Licence are tested in rotation to the extent possible.

ANNEX – B

परीक्षण उपकरणों की सूची

List Of Test Equipment

(INDICATIVE LIST, FOR GUIDANCE ONLY)

Major test equipment required to test as per the Indian Standard

Sl. No.	Tests Used In With Clause Reference	Test Equipment/Chemical
1.	Marking - 8.8	Petroleum spirit, water
2.	Dimensions – 9, General	Vernier caliper, micrometer, gauges
3.	Protection against electric shock - 10	Jointed test finger (test probe B of IS 1401), unjointed test finger (test probe 11 of IS 1401), electrical indicator, push-pull force applicator, arrangement for compression test (Figure 8), Figures 9 and 10: Gauges for checking non-accessibility of live parts, through shutters, and of live parts of socket-outlets with increased protection, Test plug
4.	Earthing Resistance - 11.5	Low voltage AC source with Voltmeter and ammeter
5.	Tests on screw terminals-12.2	Screw driver and spanner with torque meter, weights, Figure 11: Arrangement for checking damage to conductors
6.	Tests on screwless terminals Weights-12.3	Figure 11: Arrangement for checking damage to conductors, AC source, measuring instruments Figure 12: Deflection test apparatus
7.	Construction of fixed socket-outlets-13.4	Test probe 1 of IS 1401, push-pull force applicator Figure 13: Test setup
8.	Lateral strain of socket-outlets- 13.14	Figure 14: Device for checking the resistance to lateral strain
9.	Test on membranes- 13.22, 13.23	Heating cabinet, Test probe 11 of IS 1401, Freezer
10.	Test of non-solid pins- 14.2	Figure 15: Device for testing of non-solid pins, Push-pull force applicator, micrometer
11.	Temperature rise test on plugs of plug-in equipment- 14.23.1	AC source, measuring instruments, temperature measuring device
12.	Torque test on plugs of plug-in-equipment- 14.23.2	Apparatus for the torque test
13.	Ageing test- 16.1	Heating cabinet, humidity chamber

14.	Protection provided by enclosures- 16.2,	Figure 16: Test wall in accordance with the requirements of 16.2.1, Test apparatus acc. IS/IEC 60529
15.	Humidity treatment- 16.3	Humidity chamber
16.	Insulation resistance- 17.1	Insulation test equipment (Megger)
17.	Electric strength - 17.2	High voltage test equipment
18.	Temperature rise- 19	AC source, measuring instruments, temperature measuring device (thermocouples), test block for flush- mounted accessories, wooden block for surface-type socket-outlets, test plug, Clamping unit for the temperature rise test (Figure 18)
19.	Breaking capacity - 20 Normal operation- 21	Figure 19: Apparatus for breaking capacity and normal operation, test plug, fixed socket-outlet, AC source, adjustable load (resistors and inductors), measuring instruments, Figure 9: Gauge for checking non-accessibility of live parts, through shutters, after normal operation test Figure 10: Gauge for checking non-accessibility of live parts, through shutters, and of live parts of socket- outlets with increased protection
20.	Verification of the maximum withdrawal force- 22.1	Figure 21: Apparatus for checking the withdrawal force, test-plug, chemical degreaser, Test pin gauge as per Figure 22
21.	Verification of the minimum withdrawal force- 22.2	Figure 22: Gauge for the verification of the minimum withdrawal force,
22.	Test of the cord retention- 23.2	Figure 23: Apparatus for testing the cord retention, Apparatus for the torque test
23.	Flexing test- 23.4	Figure 24: Apparatus for flexing test, measuring instruments
24.	Mechanical Strength- 24.1	Pendulum hammer test apparatus, mounting arrangement, Figures 25: Mechanical strength test apparatus, Test probe B of IS 141, Steel wire of Figure 10,
25.	Tumbling barrel test- 24.2	Tumbling barrel as per Figure 26
26.	Test on ordinary surface-type socket-outlets- 24.3	Cylinder of rigid steel sheet, Flat steel sheet
27.	Impact test at low temperature- 24.4	Figure 27: Apparatus for impact test at low temperature, sponge rubber, freezer
28.	Impact test at low temperature- 10.1, 24.5	Figure 8: Arrangement for compression test
29.	Test on screwed glands-24.6	Cylindrical metal rods, Test-spanner with torque meter
30.	Abrasion test – 24.7	Figure 28: Apparatus for abrasion test on insulating sleeves of plug pins
31.	Test on shutters- 24.8	Test-pin, electrical indicator
32.	Test on multiple portable socket-outlets- 24.9	Figure 29: Arrangement for mechanical strength test on multiple portable socket-outlets

33.	Test of the fixing of the pins- 24.10	Figure 30: Arrangement to verify the fixation of pins in the body of the plug, heating cabinet
34.	Test on portable socket-outlets with suspension means- 24.11- 24.12	Cylindrical steel rod
35.	Test on portable socket-outlets with suspension means- 24.13	Round head screw
36.	Removal of covers or cover-plates Weights- 24.14-24.19	Figure 31: Arrangement for test on covers or cover- plates, Figure 32: Gauge for the verification of the outline of covers or cover plates, Figure 35: Gauge for verification of grooves, holes and reverse tapers, Figure 37: Apparatus for compression test for the verification of resistance to heat
37.	Resistance to heat – 25	Heating cabinet, Test probe B of IS 1401, Figure 38: Ball pressure test apparatus, Figure 37: Apparatus for compression test for the verification of resistance to heat
38.	Screws, current carrying parts and connections- 26	Screw-driver and spanner with torque meter
39.	Creepage distances, clearances-27	Calipers. Tolerances gauges
40.	Glow-wire test- 28.1.1	Glow wire test apparatus according to IS 11000 (Part 2/ Sec 1)
41.	Test of the resistance to heat of pins with insulating sleeves- 28.1.2	Figure 39: Apparatus for testing resistance to abnormal heat of insulating sleeves of plug pins
42.	Resistance to tracking- 28.2	Test apparatus according to IS 2824
43.	Resistance to rusting- 29	Chemicals (Ammonium Chloride), Humidity cabinet, Heating cabinet
44.	Pressure test at high temperature- 30.1	Figure 40: Apparatus for pressure test at high temperature, Heating cabinet
45.	Static damp heat test- 30.2	Climatic chamber according to IS 9000 (Part 5/ Sec 1 & 2)
46.	Test at low temperature- 30.3	Freezer
47.	Impact test at low temperature- 30.4	Figure 41: Impact test apparatus on pins provided with insulating sleeves

Note: The above list is indicative only and may not be treated as exhaustive.

ANNEX – C

निरीक्षण और परीक्षण की स्कीम

Scheme Of Inspection And Testing

1. QUALITY ASSURANCE PLAN

1.1 It is expected that manufacturers (licensees/applicants) will implement a Quality Assurance Plan i.e. a plan of regular testing and in-process controls, designed to ensure that the product bearing the Standard Mark conforms to all requirements of the Indian Standard.

1.2 The manufacturers shall define a Quality Assurance Plan defining the control unit (i.e. lot/batch etc.) and the levels of control (i.e. the frequency and number of samples for conducting the different tests as per the Indian Standard) and submit the same to BIS Branch Office for information. The manufacturer shall comply with the same and maintain test records in accordance with para 2.4.

1.3 RECOMMENDED LEVELS OF CONTROL/CONTROL UNIT:

1.3.1 For the guidance of manufacturers, the recommended definition of control unit is: **Plugs or Socket-Outlets of same type and rating manufactured in a day shall constitute a control unit.** The manufacturer has the choice to define their own control unit/batch/lot and submit the same to the BIS

1.3.2 For the guidance of manufacturers in preparing the Quality Assurance Plan, recommended levels of control are given in **Table 1**.

1.3.3 The manufacturer shall ensure inspection and testing as per the Quality Assurance Plan submitted by them on the whole production of the factory which is covered by this plan. Alternatively, the manufacturer has the option of adherence to the quality plan as per levels of control recommended in column 3 of Table 1.

1.4 However, all manufacturers shall ensure compliance of their products to all the requirements of the Indian Standard.

2. ENSURING COMPLIANCE THROUGH TESTING- It is expected that manufacturers (licensees/applicants) will establish a suitably equipped and staffed in house laboratory (In house testing facility) for testing at least those parameters of the Indian Standard which require routine testing for ensuring quality of the product. This includes in-process controls as may be defined and put in place by the manufacturer and testing parameters/requirements which can only be performed in the factory.

2.1 For the guidance of manufacturers, Table 1 giving the recommended levels of control is given below. Column 2 of Table 1 indicates routine tests where test equipment is required in house as “R” or other tests which can be subcontracted as “S”. Subcontracting is permitted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.2 For MSME manufacturers, the requirement of maintaining a laboratory/in-house testing facility for routine tests (indicated as “R” in Column 2 of Table 1) is also optional.

2.2.1 MSME manufacturers may utilize common cluster based facilities as per guidelines for the utilization of cluster based test facilities by MSMEs or the provisions of Sharing of testing facilities or get testing done from

BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.3 Large Scale manufacturers shall maintain an in-house laboratory equipped at least with test facilities for routine tests (indicated as “R” in Column 2 of Table 1), where different tests given in the specification shall be carried out in accordance with the method given in the specification. They shall also implement a calibration plan for the in-house test equipment.

2.3.1 Alternatively, in lieu of an in-house laboratory, large scale manufacturers can also utilize the provisions of Sharing of testing facilities as per the Guidelines for Grant of Licence available on BIS website www.bis.gov.in. (Under Conformity Assessment>Product Certification Process). Even for subcontracted tests, provisions for sharing of testing facilities can be utilized.

2.4 TEST RECORDS- The manufacturers maintaining an in-house laboratory or utilizing common cluster based facilities or shared test facilities shall maintain test records for the tests carried out to establish conformity. For the tests being subcontracted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025, test reports issued by the laboratories shall be available for inspection by BIS.

3. PACKING AND MARKING - The Standard Mark as given in the Schedule of the licence shall be incorporated legibly and indelibly on **each Plug or Socket-Outlet and also on its packaging** provided always that the material so marked conforms to each requirement of the specification.

3.1 Packing and Marking shall be done as per the **IS 1293 : 2019**.

3.2 **Additional Marking requirements:** The material shall also be marked with the following additional requirement on packaging:

a) “For BIS certification details please visit www.bis.gov.in”

4. REJECTION - All the production which conforms to the Indian Standard and covered under the scope of this licence shall be marked with the Standard Mark. Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1
(ONLY FOR GUIDANCE PURPOSE)

(1)				(2)	(3)		(4)
Test Details				Test equipment requirement R: required (or) S: Sub - contracting permitted	Levels of Control		Remarks
Cl.	Requirement	Test Methods			No. of Sample	Frequency	
		Clause	Ref				
8	Marking (except 8.8)	8	IS 1293	---	Firm should exercise adequate control so that whole production should meet the requirements.		
8.8	Marking (Marking shall be durable and easily legible)	8.8		R			
9	Checking of dimensions	9		R			
10	Protection against electric shock	10		R	3	Every week for each type and rating of plugs and socket-outlets manufactured in the period	
22	Force necessary to withdraw the plug	22		R	3		
17.3	Electric Strength (Flash test)	17.3		R	3	Every three month for each type and rating of plugs and socket-outlets manufactured in the period	
19	Temperature rise	19		R	3		
16	Resistance to harmful ingress of water and to humidity test	16.2, 16.3		S	3		
17	Insulation resistance and electric strength	17		S	3		
24	Mechanical Strength	24		S	3		
20	Making and breaking capacity	20		S	3		Refer Note-1
13	Constructional requirements of fixed socket outlets	13		S	3		
14	Constructional requirements of plugs and portable socket outlets	14		S	3	Once in a year for each type and rating of plugs and socket-outlets manufactured in the period	

15	Interlocked socket outlet	15		S	3	Once in a year for each type and rating of plugs and socket-outlets manufactured in the period	
23	Flexible cables and their connection	23		S	3		
25	Resistance to heat	25		S	3		
26	Screws, current carrying parts and connections	26		S	3		
27	Creepage distances, clearance and distances through sealing compound	27		S	3		
11	Provision for earthing	11		S	3		
12	Terminals	12		S	3		
16	Resistance to ageing, harmful ingress of water, humidity	16		S	3		
17	Insulation resistance and electric strength	17		S	3		
18	Operation of earthing contacts	18		S	3		
21	Normal operation	21		S	3		
29	Resistance to rusting	29		S	3	Once in a year or whenever there is change in raw material of plugs and socket-outlets manufactured in the period	
30	Additional tests on pins provided with insulating sleeves	30		S	3		
28	Resistance of insulating material to abnormal heat, fire and tracking	28		S	3		

Note-1: The number of additional specimens for testing shall be selected as per clause 5 of IS 1293. In case of any failure, the provisions given in clause 5.5 of IS 1293 shall be followed.

ANNEX – D

एक दिन में संभावित परीक्षण Possible Tests in a day

1. Marking (Cl. 8)
2. Checking of dimensions (Cl. 9)
3. Protection against electric shock (Cl. 10)
4. Electric Strength (Flash test) (Cl. 17.3)
5. Temperature rise (Cl. 19)
6. Tumbling barrel test (Cl. 24.2)
7. Creepage distance and clearances (Cl. 27)
8. Glow-wire test (Cl. 28.1.1)

ANNEX – E

लाइसेंस का दायरा **Scope of Licence**

Licence is granted to use Standard Mark as per IS 1293:2019 with the following scope:

Name of the Product	
Type (socket-outlet)	Fixed/ Portable/ multiple portable socket-outlets / Combined/ for Appliances
Rated Current	
Rated Voltage	
Nature of supply	
IP (plugs/socket outlets)	
Provision for earthing (plugs/socket outlets)	With/ without earthing contact
Method of connecting cable (plugs/socket outlets)	Rewirable/ non-rewirable
Type of terminals (plugs/socket outlets)	Screw-type/ screwless for rigid conductors/ screwless for rigid and flexible conductors
Degree of protection against electric shock (socket-outlets)	Normal / Increased protection
Method of application/ mounting (socket-outlet)	Surface/ Flush/ Semi-Flush/ Panel/ Architrave Type
Existence of shutters (socket-outlet)	With/ without shutters
Intended Use (if any) (socket-outlets)	
Intended Class of appliance (Plugs)	Class I/ Class II
Insulating sleeves (Plug)	Pins with/ without insulating sleeve

Scope for Plugs and Socket-outlets may be provided in separate tables in the Licence.

ANNEX – F

Guidelines for Outsourcing of Moulded Bodies

a) Outsourcing of moulding facilities may be permitted with the following organizations (Moulder):

For Domestic Manufacturers

- Any dedicated moulding unit in the conforming/ authorized industrial area and located under the jurisdiction of the same Branch Office.
- Any other unit of the Licensee/ Applicant or their sister concern, which is having a valid BIS licence for the same product under the jurisdiction of the same Branch Office.

For Foreign Manufacturers

- Any dedicated moulding unit in the conforming/ authorized industrial area and located in the same city where the applicant/ licensee is located.
- b) The dedicated moulder shall not carry out moulding for units other than the particular Applicant/ Licensee.
- c) The Manufacturer shall take the following actions:
- Enter into an agreement with the moulder which shall comprise, inter alia, the terms and conditions for moulding including the condition of dedicated moulding, period of agreement etc.
 - Declare the Items and Brands for which he intends to outsource moulding.
 - Declare the name, ownership details, address, details of the responsible person, phone number, etc. of the moulder.
- d) It shall be the responsibility of the licensee (the firm which is outsourcing the moulding activity) to ensure that no finished material is produced or assembled at any place other than the licensed premises and he shall be responsible for any misuse of ISI mark at the moulder's premises. An Undertaking to this effect shall be furnished to BIS.
- e) Records of raw material issued date-wise, moulded material received back date-wise and item-wise shall be maintained by the licensee. Similar records shall also be maintained by the moulder.
- f) The moulder shall ensure that there is a separate dedicated space for keeping moulded components as well as the raw materials received from the licensee. Colour coding of bins/ separate storage location etc. may be used to ensure easy and quick traceability. Appropriate records are to be maintained at both the units.
- g) The licensee shall ensure that there are adequate means of ensuring the quality of incoming outsourced moulded bodies.

- h) For applications received with request for outsourcing of moulding facilities, a special inspection of the moulder's premises shall be carried out to adjudge the capability of the moulder in addition to the preliminary inspection at the applicant's premises. Special Inspection Charges and any other charges, as applicable, for this visit shall be collected from the applicant in advance. The process flow chart submitted by the applicant shall clearly indicate the outsourcing of moulded bodies. The recommendations for GoL shall include details on outsourcing of moulding operation by the firm. The details of outsourcing may also be included in the GoL intimation letter.
- i) For existing licensees who intend to outsource the moulding facility, the licensee shall submit a fresh process flow chart indicating the outsourcing of moulded bodies. The permission for outsourcing may be given to the licensee after complying with all the procedures including verification visit to the moulder's premises.
- j) Whenever there is a change in the moulder, the licensee has to enter into a fresh agreement with the new moulding unit. The permission for outsourcing may be given to the licensee after complying with all the procedures including verification visit to the moulder's premises.
- k) During the operation of the licence, records required to be maintained at the licensee end with respect to outsourced moulding shall be verified during surveillance visit at the licensee's premises by BIS Certification Officer.
- l) BIS Officer(s) shall have access to the moulder's premises and if at any time the same is denied, permission for outsourcing shall be withdrawn. An undertaking in this respect shall be obtained.
- m) For Domestic manufacturers, the visit to moulder's premises may be taken up as a special visit as and when needed. Special Inspection Charges for this visit shall be collected from the licensee after the visit. An undertaking from the licensee regarding payment of such charges shall be obtained.
- n) For Foreign manufacturers, BIS shall maintain surveillance over the moulder's premises in addition to the manufacturer's premises. Special inspection charges and all other applicable charges for this visit shall be payable by the licensee.

Note:

- The Indian Standard does not impose any restriction on the manner/ method of Marking so long as the Durability clause is complied with. As such, the product may be marked with the Standard Mark, CM/L Number, Manufacturer's Brand and other markings during the process of moulding itself, in which case all the conditions stated above shall apply.
- However, if the Marking on the moulded bodies is done at the Licensee's own premises by other means such as pad printing/ laser marking etc, the moulded component may be treated as any other bought out component/ raw material as these are not marked with the ISI Mark or CM/L Number during the outsourced moulding process. Notwithstanding this, the Applicant/ Licensee shall ensure that there are adequate means of ensuring the quality of the incoming components.



उत्पाद मैनुअल

आई एस 3854 : 2023 के अनुसार

घरेलु और समान कार्यों के लिए स्विच - विशिष्टि (तीसरा पुनरीक्षण)के लिए

दस्तावेज संख्या - पीएम/आईएस 3854/3/ जनवरी 2024

विभिन्न उत्पादों के लिए भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम, 2018 की योजना- I के तहत प्रमाणन के संचालन में एकरूपता और पारदर्शिता के लिए इस उत्पाद मैनुअल का उपयोग सभी क्षेत्रीय / शाखा कार्यालयों और लाइसेंसधारियों द्वारा संदर्भ सामग्री के रूप में किया जाएगा। दस्तावेज का उपयोग बीआईएस प्रमाणन प्राप्त करने के इच्छुक संभावित आवेदकों द्वारा भी किया जा सकता है।

PRODUCT MANUAL FOR SWITCHES FOR DOMESTIC AND SIMILAR PURPOSES — SPECIFICATION (THIRD REVISION) ACCORDING TO IS 3854 : 2023

Document No - PM/IS 3854/3/January 2024

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure uniformity of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification.

भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, ९, बहादुर शाह ज़फ़र मार्ग
Manak Bhawan, 9, Bahadur Shah Zafar Marg
नई दिल्ली- ११०००२
New Delhi – 110002



उत्पाद मैनुअल
घरेलु और समान कार्यों के लिए स्विच - विशिष्टि (तीसरा पुनरीक्षण)
आई एस 3854 : 2023 के अनुसार

PRODUCT MANUAL
FOR SWITCHES FOR DOMESTIC AND SIMILAR
PURPOSES — SPECIFICATION (THIRD
REVISION) ACCORDING TO IS 3854 : 2023

विभिन्न उत्पादों के लिए भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम, 2018 की योजना- I के तहत प्रमाणन के संचालन में एकरूपता और पारदर्शिता के लिए इस उत्पाद मैनुअल का उपयोग सभी क्षेत्रीय / शाखा कार्यालयों और लाइसेंसधारियों द्वारा संदर्भ सामग्री के रूप में किया जाएगा। दस्तावेज़ का उपयोग बीआईएस प्रमाणन प्राप्त करने के इच्छुक संभावित आवेदकों द्वारा भी किया जा सकता है।

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure uniformity of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification.

1.	मानक संख्या IS No.	:	IS 3854 : 2023
	शीर्षक Title	:	Switches for Domestic and Similar Purposes — Specification (Third Revision)
	संशोधनों की संख्या No. of amendments	:	0
2.	नमूना दिशानिर्देश Sampling Guidelines:		
a)	कच्चा माल Raw material	:	NA
b)	समूहीकरण दिशानिर्देश Grouping Guidelines	:	Please refer ANNEX – A

c)	नमूने का परिमाण Sample Quantity	:	25 Switches
3.	परीक्षण उपकरणों की सूची List of Test Equipment	:	Please refer ANNEX – B
4.	निरीक्षण और परीक्षण की स्कीम Scheme of Inspection and Testing	:	Please refer ANNEX – C
5.	एक दिन में संभावित परीक्षण Possible tests in a day:	:	Please refer ANNEX-D
6.	लाइसेंस का दायरा /Scope of the Licence: Please refer ANNEX-E		
7.	अन्य दिशानिर्देश/ Any other guidelines: Please refer ANNEX-F for guidelines for outsourcing of moulded bodies		

ANNEX – A

समूहीकरण दिशानिर्देश Grouping Guidelines

1. The following aspects of a switch shall be taken into consideration for grouping of different varieties of switches for the purpose of GoL/ CSoL:
 - a. Current rating
 - b. Voltage rating
 - c. Possible connections or pattern
 - d. Amperes rating (rated current for externally ballasted loads)
 - e. Contact opening
 - f. Degree of protection against harmful ingress of solid foreign objects
 - g. Degree of protection against harmful ingress of water
 - h. Method of activating the switch
 - i. Method of mounting the switch
 - j. Type of terminals
 - k. Degree of protection against electric shock
2. To cover a range of current ratings, switches from the minimum and maximum rating of that range shall be tested.
3. The following relaxation may be given when a variety is tested for all the requirements:
 - a) If a particular voltage rating is tested, lower ratings may also be covered.
 - b) If a pattern is tested, additions pattern may be covered as given below:

Pattern tested	Additional Pattern
6	1
4	6,1
6/2	6,2
7	6/2, 2
03	3

- c) If switch with Amperes rating (AX) (rated current for externally ballasted loads) is tested, then separate testing may not be carried out for normal current rating (A).
 - d) If switch with a particular degree of protection is tested, switches with lower degree of protection may be covered.
4. For all other aspects mentioned at 1.1(h) to 1.1(j) above, each aspect shall be tested to cover the entire range.

5. The Firm shall declare the varieties of Switches intended to be covered in the Licence.
6. Scope of licence shall be restricted based on the manufacturing and testing facilities available.
7. During operation of licence, samples of each variety covered in the scope of licence, shall be tested in rotation, to the extent possible.

ANNEX – B

परीक्षण उपकरणों की सूची

List Of Test Equipment***(INDICATIVE LIST, FOR GUIDANCE ONLY)******Major test equipment required to test as per the Indian Standard***

Sl. No.	Tests Used In With Clause Reference	Test Equipment/Chemical
1.	Marking - 8.9	Petroleum Spirit
2.	Dimensions - 9	Vernier Calliper
3.	Protection against electric shock - 10	Un-jointed test finger, Standard test finger, Voltmeter , Voltage Source with indicator
4.	Earthing Resistance - 11.4	Low voltage source with Voltmeter and ammeter
5.	Resistance to Humidity - 15.3	Humidity Controlled Cabinet, Thermometer Hygrometer
6.	Insulation resistance - 16.1	Megger (500 V dc)
7.	Electric strength (flash test) - 16.2, 16.3	High voltage Tester (ac), Voltmeter, Ammeter
8.	Resistance to Ageing, Heat, Rusting - 15, 21, 25	Electrical Oven
9.	Temperature Rise Test, Making-Breaking and Normal Operation Test - 17, 18, 19	Temperature Rise Test Equipment, Thermometer(Digital), Ammeter, Pine wood block
10.	Terminals - 12, 13, 22	Torque Screw Driver
11.	Making-Breaking Capacity and Normal operation test – 18, 19	Test Apparatus, cycle counter, ON Timer OFF Timer, Inductive Load of required power factor, Voltmeter, Auto Transformer, PF Meter, Tungsten Filament Lamp load, Fluorescent Lamp Load
12.	Mechanical Strength – 20	Mechanical Strength (Impact test apparatus) with plywood base for mounting (Pendulum Type)
13.	Pendulum Hammer Test - 20.2	Cylinder of steel sheet and flat steel sheet (for surface type switches)

14.	Test on the Main Parts of Surface-type Switches - 20.3	Cylindrical metal rod
15.	Screwed Glands - 20.4	Setup for verification of removal and non-removal of covers
16.	Covers, Cover plates - 20.7, 20.8	Gauges as per Fig 18 to 21
17.	Grooves, Holes and Reverse Tapers - 20.9	Test setup on cord operated switch
18.	Resistance to heat – 21	Ball Pressure Apparatus (Fig 23), Magnifying glass
19.	Resistance to Rusting – 25	CCL ₄ , NH ₄ Cl, Distilled Water & Beaker
20.	Resistance to Ageing - 15.1	Weighing Balance, Weights
21.	Protection provided by enclosures of switches -15.2	Splash proof / Jet proof test setup (for IPX4/IPX5)
22.	Resistance to Abnormal Heat, Fire and Tracking - 24	Glow Wire Test Apparatus with Temperature Indicator & Controller, Timer pre-set Time Interval Indicator, Tracking test apparatus (for switches other than ordinary)
23.	Terminals & Screws - 12.2.5	Test setup for checking conductor damage with rotational speed controller and weights
24.	Screw less Terminal testing - 12.3.10, 12.3.11, 12.3.12	Test setup for mechanical stress, electrical and thermal stress, and Deflection test on Screw less terminals
25.	Attachment of knobs - 13.5	Testing arrangement for knob of rotary switches
26.	Requirements for Membranes in Inlet Openings - 13.15	Fixture for membrane reliability test
27.	Pull force test on cord operate switch - 14.6	Pull force test on cord operate switch setup
28.	Test for Switches Intended for Self-Ballasted Lamp Loads – 19.3	Load bank

Note: The above list is indicative only and may not be treated as exhaustive.

ANNEX – C

निरीक्षण और परीक्षण की स्कीम

Scheme Of Inspection And Testing

1. QUALITY ASSURANCE PLAN

1.1 It is expected that manufacturers (licensees/applicants) will implement a Quality Assurance Plan i.e. a plan of regular testing and in-process controls, designed to ensure that the product bearing the Standard Mark conforms to all requirements of the Indian Standard.

1.2 The manufacturers shall define a Quality Assurance Plan defining the control unit (i.e. lot/batch etc.) and the levels of control (i.e. the frequency and number of samples for conducting the different tests as per the Indian Standard) and submit the same to BIS Branch Office for information. The manufacturer shall comply with the same and maintain test records in accordance with para 2.4.

1.3 RECOMMENDED LEVELS OF CONTROL/CONTROL UNIT:

1.3.1 For the guidance of manufacturers, the recommended definition of control unit is : Switches of same type and rating manufactured in a day shall constitute a control unit. The manufacturer has the choice to define their own control unit/batch/lot and submit the same to the BIS

1.3.2 For the guidance of manufacturers in preparing the Quality Assurance Plan, recommended levels of control are given in **Table 1**.

1.3.3 The manufacturer shall ensure inspection and testing as per the Quality Assurance Plan submitted by them on the whole production of the factory which is covered by this plan. Alternatively, the manufacturer has the option of adherence to the quality plan as per levels of control recommended in column 3 of Table 1.

1.4 However, all manufacturers shall ensure compliance of their products to all the requirements of the Indian Standard.

2. ENSURING COMPLIANCE THROUGH TESTING- It is expected that manufacturers (licensees/applicants) will establish a suitably equipped and staffed in house laboratory (In house testing facility) for testing at least those parameters of the Indian Standard which require routine testing for ensuring quality of the product. This includes in-process controls as may be defined and put in place by the manufacturer and testing parameters/requirements which can only be performed in the factory.

2.1 For the guidance of manufacturers, Table 1 giving the recommended levels of control is given below. Column 2 of Table 1 indicates routine tests where test equipment is required in house as “R” or other tests which can be subcontracted as “S”. Subcontracting is permitted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.2 For MSME manufacturers, the requirement of maintaining a laboratory/in-house testing facility for routine tests (indicated as “R” in Column 2 of Table 1) is also optional.

2.2.1 MSME manufacturers may utilize common cluster based facilities as per guidelines for the utilization of cluster based test facilities by MSMEs or the provisions of Sharing of testing facilities or get testing done from

BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.3 Large Scale manufacturers shall maintain an in-house laboratory equipped at least with test facilities for routine tests (indicated as “R” in Column 2 of Table 1), where different tests given in the specification shall be carried out in accordance with the method given in the specification. They shall also implement a calibration plan for the in-house test equipment.

2.3.1 Alternatively, in lieu of an in-house laboratory, large scale manufacturers can also utilize the provisions of Sharing of testing facilities as per the Guidelines for Grant of Licence available on BIS website www.bis.gov.in. (Under Conformity Assessment>Product Certification Process). Even for subcontracted tests, provisions for sharing of testing facilities can be utilized.

2.4 TEST RECORDS- The manufacturers maintaining an in-house laboratory or utilizing common cluster based facilities or shared test facilities shall maintain test records for the tests carried out to establish conformity. For the tests being subcontracted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025, test reports issued by the laboratories shall be available for inspection by BIS.

3. PACKING AND MARKING - The Standard Mark as given in the Schedule of the licence shall be incorporated legibly and indelibly on **each switches and also on its packaging** provided always that the material so marked conforms to each requirement of the specification.

3.1 Packing and Marking shall be done as per the **IS 3854 : 2023**.

3.2 Additional Marking requirements: The material shall also be marked with the following additional requirement on packaging:

a) “For BIS certification details please visit www.bis.gov.in”

4. REJECTION - All the production which conforms to the Indian Standard and covered under the scope of this licence shall be marked with the Standard Mark. Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1
(ONLY FOR GUIDANCE PURPOSE)

(1)				(2)	(3)		
Test Details				Test equipment requirement R: required (or) S: Sub-contracting permitted	Levels of Control		
Cl.	Requireme nt	Test Methods			No. of Sample	Frequency	Remarks
		Claus e	Referenc e				
8	Marking (except 8.9)	8	IS 3854	R		Each Switch	
16.3	Electric Strength (Flash test)	16.3		R	3	Every week for each type and rating	
8.9	Marking – Durability	8.9		R	3	Each control unit	
9	Checking of Dimensions	9		R	3	Each control unit	Refer Note 1
10	Protection against electric shock	10		R	3	Every week for each type and rating	
11	Provision for earthing	11		S		Once in three months for each type and rating	Refer Note 2
12	Terminals	12		S			
13	Constructional requirements	13		S			
14	Mechanism	14		S			
15	Resistance to ageing, harmful ingress of water and to humidity	15		S			
16	Insulation resistance and electric strength (except 16.3)	16		S			
17	Temperature rise	17		S			
19	Normal operation	19		S			
20	Mechanical Strength	20		S			

21	Resistance to heat	21		S			
22	Screws, current-carrying parts and connections	22		S			
23	Creepage distances, clearance and distances through sealing compound	23		S			
25	Resistance to rusting	25		S			
24	Resistance to insulating material to abnormal heat, to fire and to tracking	24		S	3	Once in a year or whenever there is change in raw material for each type and rating	Refer Note 2
18	Making and breaking capacity	18		S	3	Once in three years for each type and rating for each type and rating	Refer Note 2

Note-1: Firm should exercise adequate control so that whole production should meet the requirements. Records shall be maintained.

Note-2: The number of specimens for testing shall be as per selected as per clause 5.3 and Table 1 of IS 3854. In case of any failure, the provisions given in clause 5.3 of IS 3854 shall be followed.

Note-3: Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empaneled by the Bureau

Note-4: Levels of control given in table above are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control with proper justification for approval by BO Head.

ANNEX – D

एक दिन में संभावित परीक्षण Possible Tests in a day

1. Marking (Cl. 8)
2. Protection against electric shock (Cl. 10)
3. Electric Strength (Flash test) (Cl. 16.3)
4. Temperature rise (Cl. 17)
5. Making and breaking capacity – General (Cl. 18.1)
6. Mechanical Strength (Cl. 20)
7. Resistance to heat (Cl. 21)
8. Screws, current-carrying parts and connections (Cl. 22)
9. Creepage distances, clearance and distances through sealing compound (Cl. 23)
10. Resistance to rusting (Cl. 25)

ANNEX – E

लाइसेंस का दायरा
Scope of Licence

Licence is granted to use Standard Mark as per IS 3854:2023 with the following scope:

Name of the Product	Switches for domestic and similar purposes
Rated Current	
Rated Voltage	
Nature of supply	
Possible Connections (PatternNumber)	
Amperes rating (rated current for externally ballasted loads)	AX/A
Contact opening	Normal gap construction / mini-gap Construction/ micro-gap construction/ switches without contact gap (semiconductor switching devices)
Degree of protection against Electric Shock	Unenclosed / Enclosed
Degree of protection against harmful ingress of Water and against harmful ingress of solid foreign objects	IP....
Method of activating the switch	Rotary/ Tumbler/ Rocker/ Push-Button/ Cord-operated/ Dolly type
Method of Mounting	Surface-type/ Flush-type/ Semi flush type/ Panel-type/ Architrave-type
Type of Terminals	Screw-type terminals/ Screwless terminals for rigid conductors only/ Screwless terminals for rigid and flexible conductors only

ANNEX – F

Guidelines for Outsourcing of Moulded Bodies

a) Outsourcing of moulding facilities may be permitted with the following organizations (Moulder):

For Domestic Manufacturers

- Any dedicated moulding unit in the conforming/ authorized industrial area and located under the jurisdiction of the same Branch Office.
- Any other unit of the Licensee/ Applicant or their sister concern, which is having a valid BIS licence for the same product under the jurisdiction of the same Branch Office.

For Foreign Manufacturers

- Any dedicated moulding unit in the conforming/ authorized industrial area and located in the same city where the applicant/ licensee is located.
- b) The dedicated moulder shall not carry out moulding for units other than the particular Applicant/ Licensee.
- c) The Manufacturer shall take the following actions:
- Enter into an agreement with the moulder which shall comprise, inter alia, the terms and conditions for moulding including the condition of dedicated moulding, period of agreement etc.
 - Declare the Items and Brands for which he intends to outsource moulding.
 - Declare the name, ownership details, address, details of the responsible person, phone number, etc. of the moulder.
- d) It shall be the responsibility of the licensee (the firm which is outsourcing the moulding activity) to ensure that no finished material is produced or assembled at any place other than the licensed premises and he shall be responsible for any misuse of ISI mark at the moulder's premises. An Undertaking to this effect shall be furnished to BIS.
- e) Records of raw material issued date-wise, moulded material received back date-wise and item-wise shall be maintained by the licensee. Similar records shall also be maintained by the moulder.
- f) The moulder shall ensure that there is a separate dedicated space for keeping moulded components as well as the raw materials received from the licensee. Colour coding of bins/ separate storage location etc. may be used to ensure easy and quick traceability. Appropriate records are to be maintained at both the units.
- g) The licensee shall ensure that there are adequate means of ensuring the quality of incoming outsourced moulded bodies.

- h) For applications received with request for outsourcing of moulding facilities, a special inspection of the moulder's premises shall be carried out to adjudge the capability of the moulder in addition to the preliminary inspection at the applicant's premises. Special Inspection Charges and any other charges, as applicable, for this visit shall be collected from the applicant in advance. The process flow chart submitted by the applicant shall clearly indicate the outsourcing of moulded bodies. The recommendations for GoL shall include details on outsourcing of moulding operation by the firm. The details of outsourcing may also be included in the GoL intimation letter.
- i) For existing licensees who intend to outsource the moulding facility, the licensee shall submit a fresh process flow chart indicating the outsourcing of moulded bodies. The permission for outsourcing may be given to the licensee after complying with all the procedures including verification visit to the moulder's premises.
- j) Whenever there is a change in the moulder, the licensee has to enter into a fresh agreement with the new moulding unit. The permission for outsourcing may be given to the licensee after complying with all the procedures including verification visit to the moulder's premises.
- k) During the operation of the licence, records required to be maintained at the licensee end with respect to outsourced moulding shall be verified during surveillance visit at the licensee's premises by BIS Certification Officer.
- l) BIS Officer(s) shall have access to the moulder's premises and if at any time the same is denied, permission for outsourcing shall be withdrawn. An undertaking in this respect shall be obtained.
- m) For Domestic manufacturers, the visit to moulder's premises may be taken up as a special visit as and when needed. Special Inspection Charges for this visit shall be collected from the licensee after the visit. An undertaking from the licensee regarding payment of such charges shall be obtained.
- n) For Foreign manufacturers, BIS shall maintain surveillance over the moulder's premises in addition to the manufacturer's premises. Special inspection charges and all other applicable charges for this visit shall be payable by the licensee.

Note:

- The Indian Standard does not impose any restriction on the manner/ method of Marking so long as the Durability clause is complied with. As such, the product may be marked with the Standard Mark, CM/L Number, Manufacturer's Brand and other markings during the process of moulding itself, in which case all the conditions stated above shall apply.
- However, if the Marking on the moulded bodies is done at the Licensee's own premises by other means such as pad printing/ laser marking etc, the moulded component may be treated as any other bought out component/ raw material as these are not marked with the ISI Mark or CM/L Number during the outsourced moulding process. Notwithstanding this, the Applicant/ Licensee shall ensure that there are adequate means of ensuring the quality of the incoming components.